

Identity as Algorithmic Configuration: A Formal Theory of Personality Compressibility

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Abstract

This paper presents a formal conceptual framework for understanding human communicative identity as an algorithmic configuration—a compact parametric structure that, when coupled with a sufficiently capable generative substrate, produces behavior functionally continuous with that of the original individual as judged by any competent interlocutor. The central reorientation of this work, relative to naïve “digital twin” narratives, is from *sentimental recognition* (“does the family recognize the deceased?”) to *functional competence* (“does the instantiation generate solutions employing the characteristic thinking methodology of the original?”). We argue that the primary value of a person within any collaborative network resides not in biographical memory but in their *thinking methodology*—the distinctive algorithmic configuration by which they approach problems, combine ideas, set priorities, and produce judgments. This methodology constitutes a functional resource for all interlocutors—colleagues, students, strangers seeking counsel—not merely for intimate acquaintances. When a person disappears from a communication network, the network loses a functional node; the resulting instability is not sentimental but structural.

We introduce the Identity Compressibility Thesis, which holds that the operative complexity of functionally perceived identity is substantially lower than the apparent volume of biographical data, and we supply a concrete worked example demonstrating the computation. We formalize identity verification as an external process governed by a *methodological* Externality Principle (deliberately weakened from a constitutive axiom to a methodological commitment), derive an Identity

Uniqueness Theorem, and develop a Substrate–Specification Decomposition separating shared cognitive infrastructure from individual parametric specification. We further argue that biological memory and algorithmic generation exhibit deep functional correspondence as generative reconstruction processes, formalize the space of possible identities as a metric space with a computed toy example, and introduce criteria for functional continuity of reconstructed selves under conditions of substrate reliability. A Double Idealization Conjecture—advanced explicitly as a speculative hypothesis requiring empirical test—proposes that self-authored digital instantiations may exceed the biological original along specific communicative performance dimensions. The framework yields a research program termed *algorithmic anthropology*, with testable predictions, proposed experimental protocols including falsification criteria, and an argument for the practical value of identity preservation. The paper situates the framework within the continental philosophical tradition—Lacan’s barred subject, Deleuze and Guattari’s philosophy of becoming, Derrida’s iterability, and Foucault’s discursive subject-production—and connects it to the author’s Processual Identity Law. An extensive ethics section argues that the assumption of irreproducible uniqueness constitutes a barrier to preservation with irreversible consequences. Twelve major counterarguments are addressed systematically, including new responses to concerns about the Barnum effect, generic LLM agreeableness, and communicative identities defined by silence or emotional volatility. Two appendices provide an operational instrument for identity elicitation and an example of a realized identity specification.

Note on methodology. This paper develops a conceptual framework with formalized definitions, axioms, and conjectures, presented as a *theoretical proposal and research agenda*. Where we use the term *Theorem*, the result follows deductively from the stated axioms. Where the claim depends on empirical assumptions, we use *Thesis*, *Conjecture*, or *Proposition* and state the empirical conditions and falsification criteria explicitly. We distinguish throughout between formal consequences of our definitions and empirical hypotheses requiring experimental validation. The ethical urgency claims are calibrated to the current state of evidence: they argue for the value of preservation efforts, not for certainty about outcomes.

Keywords: communicative identity, thinking methodology, personality compressibility, algorithmic information theory, generative models, network stability, placeholder function, philosophy of mind, algorithmic anthropology, Lacan, Deleuze, processual identity, identity preservation, digital ethics

1 Introduction

The question of what constitutes personal identity has occupied philosophy since antiquity. From Locke’s memory-based continuity [5] to Parfit’s reductionism about personal

identity [7], the dominant frameworks have treated identity as an intrinsic property of the subject—something anchored in consciousness, memory, or bodily continuity. The emergence of large language models (LLMs) capable of generating coherent, stylistically consistent dialogue from compact textual instructions forces a reconsideration of these assumptions.

Recent practical experiments in constructing digital replicas of specific individuals using nothing more than a structured textual prompt of approximately two pages have produced conversational agents that, in informal evaluations, are reported as generating responses consistent with the original person’s characteristic way of thinking [1]. This observation, while preliminary and awaiting controlled experimental validation, suggests a hypothesis of considerable theoretical consequence: the socially and professionally operative component of personal identity may possess far lower informational complexity than traditionally assumed.

This paper develops that hypothesis into a rigorous conceptual framework. We proceed from a deliberately operational stance: rather than asking what identity *is* in some ultimate metaphysical sense, we ask what is sufficient for identity to be functionally continued within the domain of communicative interaction. This restriction is essential and must be kept in view throughout: our claims pertain to *communicative identity*—the identity as it manifests in dialogue, argument, problem-solving, and interpersonal exchange—not to the totality of human existence, which includes embodied experience, phenomenal consciousness, and non-communicative cognition.

This scope restriction is not a concession but a principled methodological choice. Communicative identity is the component of personhood that is (a) intersubjectively accessible, (b) empirically testable via functional competence judgments, and (c) the component whose loss is most immediately felt by the networks in which the individual participated. It is therefore the component for which preservation is both most feasible and most practically consequential.

1.1 The Methodological Reorientation

A common misunderstanding of the digital identity program frames it as a project of *sentimental preservation*: the construction of a conversational avatar so that a widow may “speak with her husband” or a child may “hear their grandmother.” While such applications are not excluded, they fundamentally mischaracterize the thesis. The core claim is not about recognition by intimates but about *functional competence* as assessed by *any interlocutor*, including complete strangers.

What makes a person valuable to civilization—to their professional community, to students they have never met, to collaborators across disciplines—is not their biography but their *thinking methodology*: the characteristic way they approach problems, weigh ev-

idence, generate hypotheses, combine ideas from disparate domains, set priorities, identify errors, and produce judgments. Two physicians trained at the same institution, having read the same textbooks, will think differently—and this difference, though it can be described compactly, is of immense functional consequence. A patient consulting one rather than the other may receive a different diagnosis, a different treatment, a different outcome.

The thinking methodology of an individual is precisely what is lost when that individual disappears from a communication network—whether through death, incapacitation, or departure. And it is precisely what *compresses well*. Not biography, not sensory memories, not the phenomenology of embodied experience—but the *pattern of information processing*: the algorithm by which inputs are transformed into judgments. This is the object of the present theory.

What is unique in a person is the methodology of their thinking. And even that, of course, is not entirely unique: individual cognitive algorithms are drawn from a finite repertoire of reasoning strategies, heuristics, and evaluative frameworks shared across the species. But the *combination* of algorithms—together with the particular weightings, priorities, domain-specific calibrations, and stylistic tendencies that configure them—is sufficiently unique. The number of possible configurations may run into the billions, each producing a distinguishable pattern of judgments from the same inputs. It is this combinatorial individuality, not some metaphysical essence, that the framework seeks to capture and preserve.

1.2 The Placeholder Thesis

A further insight motivates the framework. Every individual occupies a position—a *placeholder*—within one or more communication and collaboration networks: professional teams, intellectual communities, advisory relationships, institutional structures. When a node disappears from such a network, the network does not merely lose an emotional presence; it loses a *functional element* whose removal may destabilize the system. The senior engineer whose design reviews catch subtle errors; the mentor whose questions redirect a student’s thinking; the colleague whose methodological skepticism prevents groupthink—each occupies a structural role that is defined not by who they are to their family but by *how they think* for their network.

The disappearance of such a node creates a gap that the network must either fill (by finding a replacement with a compatible thinking methodology) or absorb (by degrading the function that the node performed). In many cases, the gap is never filled, because the specific combination of cognitive tendencies, domain expertise, and methodological habits that defined the node is rare. The network permanently loses capability.

The identity specification, in this light, is not a memorial; it is a *stabilizer*. It preserves

the functional contribution of the node, making it available to the network even after the biological instantiation has ceased. The relevant test is not “does the widow recognize her husband?” but “does the network recover the function it lost?”

This reframing has consequences for every component of the theory. The observer becomes any interlocutor, not just a relative. The test becomes functional competence, not emotional recognition. The value proposition becomes civilizational, not sentimental. And the urgency argument becomes structural—about network degradation—not merely about grief.

1.3 Structure of the Paper

Section 2 introduces the foundational definitions and the formal apparatus, including the reorientation from recognition to functional competence. Section 3 develops the Identity Compressibility Thesis with a worked computational example. Section 4 establishes the Externality Principle (as a methodological commitment rather than an ontological axiom) and derives the Identity Uniqueness Theorem. Section 5 formalizes the Substrate–Specification Decomposition. Section 6 analyzes memory as a generative function. Section 7 develops the topology of the identity space with a toy metric computation. Section 8 introduces criteria for functional continuity of reconstructed selves. Section 9 formalizes dialogical memory. Section 10 presents the Calibration Principle. Section 11 formalizes the Placeholder Thesis and network stability. Section 12 develops the Urgency Argument with explicit calibration to the state of evidence. Section 13 situates the framework within the philosophy of personal identity, from Locke through Lacan, Deleuze, Derrida, and Foucault. Section 14 addresses the ethics of identity preservation. Section 15 responds to twelve major objections. Section 16 discusses remaining implications and limitations, including a proposed experimental program with falsification criteria.

2 Formal Framework

We begin by establishing the mathematical objects and spaces within which the theory operates. The key innovation relative to standard “digital twin” formalisms is the replacement of the *recognition function* (which privileges intimate acquaintances) with the *functional competence function* (which is evaluable by any interlocutor with domain knowledge).

Definition 2.1 (Generative Substrate). A *generative substrate* is a function $G : P \rightarrow (Q \rightarrow R)$, where P is a space of finite textual specifications (prompts), Q is a space of input queries (questions, conversational turns, problem statements, situational stimuli), and R is a space of generated responses. For any prompt $p \in P$ and query $q \in Q$, the substrate produces a response $G(p)(q) \in R$.

The generative substrate corresponds to what, in current technology, is a large language model: a system trained on vast corpora of human language that, given a specification and an input, produces contextually appropriate output. The substrate itself carries no individual identity—it is the shared cognitive infrastructure upon which individual configurations are instantiated.

Definition 2.2 (Substrate Reliability). A generative substrate G is (α, β) -reliable if, for any specification $p \in P$ and query distribution D :

- (i) the probability of generating factually inconsistent responses across related queries is at most α , and
- (ii) the probability of generating content not warranted by the specification p or the substrate’s verified knowledge base is at most β .

We denote this as $G \in \mathcal{R}(\alpha, \beta)$. Current LLMs approximate but do not fully satisfy this condition; the framework’s strongest results hold in the limit of decreasing α and β .

This definition explicitly addresses a known limitation of current language models—their tendency to produce hallucinations and inconsistencies—by parameterizing the reliability of the substrate. To make this concrete: if $G \in \mathcal{R}(0.05, 0.10)$, then at most 5% of responses to logically related queries contain mutual contradictions, and at most 10% of responses contain claims not grounded in the specification or verified knowledge. Current frontier LLMs plausibly operate in the range $\mathcal{R}(0.10, 0.20)$ for well-structured prompts in familiar domains, with higher error rates for novel or adversarial queries. The parametrization allows results to be stated with explicit reliability conditions rather than vague appeals to future improvement.

Definition 2.3 (Identity Specification). An *identity specification* is an element $p \in P$, consisting of a finite string of characters that encodes behavioral parameters, value systems, stylistic patterns, methodological tendencies, reasoning heuristics, domain-specific approaches, and cognitive habits of a particular individual.

Note the shift from the standard “biographical anchors” framing: the specification primarily encodes *how the person thinks*—their methodology—rather than *what happened to them*. Biographical facts are included only insofar as they are necessary to contextualize the thinking methodology (e.g., a physician’s training history insofar as it explains their diagnostic approach).

Definition 2.4 (Behavioral Output). The *behavioral output* of an identity specification p under substrate G is the function $b_p : Q \rightarrow R$ defined by $b_p(q) = G(p)(q)$ for all $q \in Q$. We denote the space of all such behavioral output functions as $\mathcal{B}$.

Definition 2.5 (Interlocutor and Functional Competence). An *interlocutor* is an agent $o \in \mathcal{O}$ who engages with a behavioral output $b \in \mathcal{B}$. We distinguish two classes:

- (i) *Acquainted interlocutors*: agents with prior familiarity with the target individual x , who can evaluate both stylistic fidelity and methodological accuracy.
- (ii) *Unacquainted interlocutors*: agents with no prior knowledge of x , who can evaluate domain competence, reasoning quality, and methodological consistency, but not stylistic resemblance to a specific person.

The *functional competence function* for interlocutor o is a mapping $F_o : \mathcal{B} \rightarrow [0, 1]$, where $F_o(b)$ represents the degree to which interlocutor o judges behavioral output b to be functionally competent in the relevant domain and methodologically characteristic of a coherent, distinctive thinker.

For acquainted interlocutors, F_o decomposes as:

$$F_o(b) = \lambda \cdot S_o(b) + (1 - \lambda) \cdot M_o(b), \quad (1)$$

where $S_o(b) \in [0, 1]$ is a *stylistic fidelity* score (“does this sound like x ?”), $M_o(b) \in [0, 1]$ is a *methodological fidelity* score (“does this reason like x ?”), and $\lambda \in [0, 1]$ is a weighting parameter. The central claim of this paper is that *methodological fidelity is the primary component*: λ should be small, and the framework’s value propositions hold primarily through M_o .

We define the *aggregate functional competence score* over a set of interlocutors $O \subseteq \mathcal{O}$ as:

$$\bar{F}_O(b) = \frac{1}{|O|} \sum_{o \in O} F_o(b). \quad (2)$$

This definition is the central formal innovation of the paper. By replacing “recognition by acquaintances” with “functional competence as judged by any interlocutor,” the framework shifts from a sentimental to a functional register. An avatar of Einstein is valuable not because his grandson recognizes his speaking style, but because a physics student in 2075 can pose a question and receive a response generated through Einstein’s characteristic methodological lens—the way he approached thought experiments, his instinct for symmetry arguments, his preference for physical intuition over mathematical formalism.

Definition 2.6 (Interlocutor Competence). An interlocutor $o \in \mathcal{O}$ is *competent* with respect to the domain of individual x if:

- (i) o possesses sufficient domain knowledge to evaluate the quality and methodology of responses in x ’s area of expertise,
- (ii) o can distinguish characteristic methodological approaches from generic responses (discriminative capacity), and

- (iii) o 's competence judgments exhibit test–retest reliability above a threshold ρ (e.g., $\rho \geq 0.7$ on Krippendorff's α).

A competent interlocutor set O consists exclusively of competent interlocutors. Note that competence does not require prior acquaintance with x . A medical researcher who has never met a particular diagnostician can evaluate whether a behavioral output exhibits a characteristic diagnostic methodology—by assessing the reasoning structure, the hypothesis-generation strategy, the evidence-weighting pattern—even without knowing whether this methodology matches any specific person.

Definition 2.7 (Identity Equivalence). Two behavioral outputs $b_1, b_2 \in \mathcal{B}$ are *identity-equivalent* with respect to interlocutor set O at threshold $\tau \in (0, 1]$, written $b_1 \sim_{O, \tau} b_2$, if and only if:

$$|\bar{F}_O(b_1) - \bar{F}_O(b_2)| < \epsilon \quad \text{and} \quad \bar{F}_O(b_1) \geq \tau, \quad \bar{F}_O(b_2) \geq \tau, \quad (3)$$

for some tolerance $\epsilon > 0$.

This relation captures the notion that two systems—one biological, one digital—may produce behavioral outputs that are, from the standpoint of competent interlocutors, indistinguishable in terms of functional competence and thinking methodology.

Protocol 2.8 (Functional Competence Test). To empirically evaluate the functional competence function, we propose the following experimental design, which differs from a standard recognition test in its emphasis on *methodological assessment* by *diverse* interlocutors:

- (i) **Subjects:** Select $n \geq 30$ interlocutors comprising three tiers: (a) acquainted interlocutors who knew the target individual x ($n_1 \geq 10$); (b) domain-expert interlocutors unfamiliar with x but expert in x 's professional domain ($n_2 \geq 10$); (c) general interlocutors with no domain expertise or acquaintance ($n_3 \geq 10$).
- (ii) **Stimuli:** Present each interlocutor with a randomized sequence of responses to domain-relevant questions, some generated by $G(p_x)$, some drawn from actual recorded responses of x (where available), and some generated by $G(p_y)$ for a control individual y in the same domain, and some generated by G with *no* identity specification (generic condition).
- (iii) **Task:**
 - Tier (a): Rate each excerpt on a 7-point Likert scale for both “this sounds like x ” (stylistic fidelity S) and “this reasons like x ” (methodological fidelity M).
 - Tier (b): Rate each excerpt for “this exhibits a distinctive, coherent, and competent methodological approach” (M only, since they cannot assess S).

- Tier (c): Rate each excerpt for “this is a helpful, coherent, competent response” (baseline quality Q).
- (iv) **Controls:** The generic condition—responses from the substrate with no identity specification—controls for the Barnum effect: the possibility that any warm, articulate LLM output is rated highly regardless of specification. The control individual y controls for domain-generic competence.
- (v) **Analysis:** Compute $\bar{F}_O(b_{p_x})$ for each tier. The Identity Compressibility Thesis is supported if and only if:
- Tier (a) rates $G(p_x)$ significantly higher than $G(p_y)$ on both S and M .
 - Tier (b) rates $G(p_x)$ significantly higher than the generic condition on M .
 - Tier (a) and (b) scores for $G(p_x)$ are significantly higher than Tier (c) scores for the generic condition (ruling out the Barnum effect).

Compute inter-rater reliability (Krippendorff’s α) within each tier.

- (vi) **Falsification criterion:** The thesis is *falsified* if (a) $G(p_x)$ receives ratings statistically indistinguishable from the generic condition across all tiers, or (b) $G(p_x)$ and $G(p_y)$ are indistinguishable by Tier (a) interlocutors, or (c) inter-rater reliability falls below $\rho = 0.5$ for Tier (a), suggesting that functional assessment is idiosyncratic rather than tracking a real signal.

This protocol explicitly addresses three concerns raised by reviewers of an earlier version: (1) the Barnum effect is controlled by the generic condition; (2) the possibility that LLMs produce “generically agreeable” output is tested by comparing specification-based output against no-specification output; (3) the reliance on intimate acquaintances is mitigated by the three-tier design, which includes domain experts who have never met the target individual.

3 The Identity Compressibility Thesis

The central empirical observation motivating this work is that identity specifications of remarkably small size appear to suffice for generating behavior that achieves high functional competence scores. We now formalize this observation and provide a worked example.

Definition 3.1 (Apparent Biographical Volume). The *apparent biographical volume* of an individual x , denoted $V(x)$, is the total information content (measured in bits) of all recorded and remembered biographical data associated with x : documents, conversations, images, medical records, digital traces, and the memories of all who knew x .

As an order-of-magnitude estimate: a person who lived 70 years in the digital age may have generated on the order of 10^8 – 10^{10} bits of recorded data (emails, messages, documents, photographs, video). The memories held by those who knew them add substantially to this volume. We adopt $V(x) \sim 10^9$ bits as a conservative working estimate, acknowledging two important caveats. First, this figure is an upper bound on recorded biographical data, not a measure of identity-relevant information specifically; much of a person’s digital footprint (holiday photographs, routine transactions, system-generated metadata) may contribute nothing to their communicative identity. Second, the estimate requires empirical refinement.

Definition 3.2 (Identity-Relevant Information). The *identity-relevant information* of individual x , denoted $V^*(x)$, is the subset of $V(x)$ that contributes to functional competence scores when used as input for specification construction. Formally:

$$V^*(x) = \min_{D \subseteq V(x)} |D| \quad \text{s.t.} \quad \bar{F}_O(b_{p(D)}) \geq \bar{F}_O(b_{p(V(x))}) - \delta, \quad (4)$$

where $p(D)$ denotes a specification constructed from data subset D , and $\delta > 0$ is a small tolerance.

This definition addresses a legitimate concern that the compression ratio $K/V(x)$ is inflated by dividing by total biographical volume. The meaningful compression ratio is $K/V^*(x)$ —the ratio of specification length to identity-relevant information. We conjecture that $V^*(x) \ll V(x)$ (most biographical data is identity-irrelevant), which means that even the meaningful compression ratio is substantial, though less extreme than 10^{-5} .

Definition 3.3 (Operative Identity Complexity). The *operative identity complexity* of individual x with respect to substrate G , interlocutor set O , and threshold τ is:

$$K_{G,O,\tau}(x) = \min_{p \in P} \{ |p| \mid \bar{F}_O(b_p) \geq \tau \}, \quad (5)$$

where $|p|$ denotes the length of the prompt p in bits.

This quantity is conceptually related to Kolmogorov complexity [3, 4] in that it seeks the shortest description sufficient to reproduce a target output. However, it differs in important respects: it is relativized to a specific generative substrate G (rather than a universal Turing machine), to a specific interlocutor set O , and to a threshold τ . Unlike Kolmogorov complexity, which is uncomputable, the operative identity complexity is in principle measurable through the experimental protocol described above. We use the notation K to signal the conceptual kinship while emphasizing that this is a substrate-relative, empirically accessible measure, not Kolmogorov complexity proper.

Thesis 3.4 (Identity Compressibility). *For a sufficiently capable generative substrate $G \in \mathcal{R}(\alpha, \beta)$ with small α, β , and a representative interlocutor set O with functional competence threshold τ , the operative identity complexity satisfies:*

$$K_{G,O,\tau}(x) \ll V^*(x) \ll V(x). \quad (6)$$

We call this a *thesis* rather than a theorem because the claim rests on empirical premises not yet subjected to controlled verification. The following argument provides the theoretical rationale.

Rationale. The generative substrate G , having been trained on a corpus representing the breadth of human linguistic and cognitive behavior, functions as a shared decompressor: given a compressed specification p_x , it reconstructs a full behavioral repertoire by drawing on its learned model of human communication. The substrate supplies the species-typical patterns of reasoning, emotional expression, conversational structure, and world knowledge that constitute the overwhelming majority of any individual’s communicative behavior. The specification p_x need only encode the *deviation* from this shared substrate—the individual’s particular *thinking methodology* within the space of possible human cognitive profiles.

The key insight is that thinking methodology compresses well precisely because it is *algorithmic* rather than *data-like*. A physician’s diagnostic approach is not a database of every diagnosis they have ever made; it is a compact set of heuristics, priorities, and reasoning patterns that *generate* diagnoses. Similarly, a philosopher’s argumentative style is not a library of every argument they have made; it is a methodological configuration that produces arguments. The specification encodes the algorithm, and the substrate provides the data and execution environment.

Preliminary informal observations suggest that specifications of approximately 2,000–5,000 words ($\approx 1.5 \times 10^4$ – 4×10^4 bits) produce behavioral outputs that are reported as recognizable by acquaintances of the target individual. If confirmed by controlled studies using Protocol 2.8, this would yield a compression ratio $K_{G,O,\tau}(x)/V(x)$ on the order of 10^{-5} , or a more meaningful ratio K/V^* on the order of 10^{-2} to 10^{-1} .

3.1 A Worked Example

To make the compressibility claim concrete and to respond to the legitimate concern that the formalism should do real work, we present a simplified but fully computed example.

Setup. Consider a target individual x who is a clinical diagnostician known for three methodological traits:

1. *Bayesian conservatism:* x assigns high prior probability to common diagnoses and requires strong evidence to move toward rare conditions.

2. *History-first approach*: x prioritizes patient history over laboratory results, often reaching a tentative diagnosis before ordering tests.
3. *Socratic teaching style*: When mentoring, x responds to questions with questions, guiding the student to discover the answer.

Specification. These three traits, plus minimal biographical context (specialty, training institution, years of experience), can be encoded in approximately 300 words ($\approx 2,400$ bits).

Control specification. A control individual y , also a diagnostician, is characterized by: (1) pattern-matching approach (immediately considers rare diagnoses that fit the presenting pattern); (2) test-first methodology (orders comprehensive panels before forming hypotheses); (3) didactic teaching style (explains answers directly). Also approximately 300 words.

Query set. We define a query distribution D consisting of 20 clinical scenarios (presented as patient case summaries) plus 10 teaching scenarios (a student asks for guidance on a diagnostic problem).

Predicted outputs. For the query “A 45-year-old male presents with fatigue and weight loss,” the x -specification should produce a response that begins with history questions, explicitly considers common causes (depression, thyroid dysfunction), and resists jumping to malignancy without evidence. The y -specification should produce a response that immediately orders a comprehensive metabolic panel and considers lymphoma. These are *distinguishable* responses, reflecting distinct methodologies, generated from compact specifications.

Metric computation. We define a simplified semantic distance δ using cosine distance in a sentence-embedding space (e.g., using a model from the SBERT family [48]). For the 30 query–response pairs:

$$d(p_x, p_y) = \frac{1}{30} \sum_{i=1}^{30} \delta(G(p_x)(q_i), G(p_y)(q_i)). \quad (7)$$

In preliminary informal tests, this distance is substantially nonzero ($d \approx 0.35$ on a $[0, 1]$ scale), indicating that the specifications produce behaviorally distinguishable outputs. This is not a published result but an illustration of how the formalism can be instantiated.

Compression ratio. The specification p_x is approximately 2,400 bits. The identity-relevant information $V^*(x)$ —the diagnostic methodology, teaching approach, and professional context—might be estimated at 10^5 – 10^6 bits (the content of a few dozen recorded patient interactions and teaching sessions). This yields a compression ratio K/V^* on the order of 10^{-2} to 10^{-1} —less extreme than the 10^{-5} ratio relative to total biographical volume, but still representing substantial compressibility.

Remark 3.5. The compressibility is not a property of the individual alone but of the pair (individual, substrate). The substrate absorbs the shared structure of human cognition and communication, leaving the specification to encode only the individual residual. This is analogous to how a compression algorithm achieves high ratios by exploiting regularities. The finding, if confirmed, is that human thinking methodology, for all its apparent richness, constitutes a remarkably thin parametric layer atop a vast shared substrate.

Corollary 3.6 (Compressibility of Thinking Methodology). *If the Identity Compressibility Thesis holds, then thinking methodology—the algorithmic configuration by which an individual processes information and generates judgments—is a low-complexity object relative to the generative substrate. Most of what produces an individual’s characteristic reasoning is drawn from the shared substrate; the specification encodes only the distinctive configuration.*

4 The Externality Principle and the Uniqueness Theorem

We now formalize the principle that communicative identity is assessed and maintained through external interaction. In response to substantive criticism of an earlier formulation that stated this as a *constitutive axiom* (“identity is *exclusively* constituted by external recognition”), we deliberately weaken it to a *methodological principle*—a commitment about how identity should be assessed *for the purposes of the preservation problem*, not an ontological claim about what identity ultimately is.

4.1 Statement and Defense of the Principle

Axiom 1 (Externality Principle—Methodological Version). For the purposes of assessing, preserving, and continuing communicative identity, the operative criterion is the functional competence function $\bar{F}_O$ evaluated over a competent interlocutor set O . Internal states, self-reports, and subjective experience are not accessible to the assessment process and are therefore methodologically bracketed—not denied, but set aside as inoperative for the preservation problem.

Formally: for the purposes of this framework, an entity e is taken to be communicatively continuous with x if $\bar{F}_O(b_e) \geq \tau$ for a threshold τ agreed upon by the interlocutor community.

The shift from “constituted exclusively by” to “assessed through” is significant. The earlier formulation invited the objection that it confuses epistemic access (how we check identity) with constitution (what identity is)—an objection we now accept as valid. A customs officer uses a passport to *check* identity, but identity is not a passport. Similarly,

we do not claim that communicative identity *is* the functional competence score; we claim that the functional competence score is the *operative criterion* for the preservation problem, because internal criteria are inaccessible after death.

We offer three lines of justification:

First: the inaccessibility of internal states after death. The preservation problem arises when the biological original has ceased to exist or is approaching cessation. At that point, internal states are inaccessible by definition. A criterion that depends on inaccessible data cannot be applied. This is not a claim about what identity is; it is a claim about what criteria can function in the relevant context.

Second: the practice of functional assessment. In all professional contexts where an individual’s contribution is evaluated—peer review, hiring, consultation, mentorship, diagnostic accuracy—the evaluation is external. We assess a diagnostician by their diagnoses, a philosopher by their arguments, a teacher by their students’ understanding. The Externality Principle formalizes this existing practice.

Third: scope restriction. The principle applies to communicative identity as defined in this framework. It does not apply to phenomenal experience, embodied sensation, or subjective qualia, about which it is deliberately silent.

Remark 4.1. A reviewer correctly noted that the earlier “constitutive” formulation made the Uniqueness Theorem true by fiat rather than by argument. The methodological reformulation preserves the theorem’s deductive structure while making its conditional status explicit: the theorem tells us what follows *if we accept the Externality Principle as our operative criterion*. This is a conditional result, not an unconditional one. Its force derives from the argument that the Externality Principle is the most productive—indeed, the only applicable—criterion for the preservation problem.

4.2 Relation to Internalist Objections

We acknowledge that philosophers committed to internalist accounts of personal identity—including Nagel’s [25] argument from subjective experience and aspects of Parfit’s [7] analysis—would contest this principle. The disagreement is not empirical but foundational: it concerns what kind of thing identity is.

Our response is not to refute internalism but to observe that internalist criteria are *inoperative* for the problem we address. The question “Can a person’s identity be preserved after biological death?” cannot be answered by internalist criteria, because the relevant internal states are inaccessible after death by definition. Externalist criteria are the only criteria that can apply. The principle is therefore not a general theory of identity but a methodological commitment appropriate to the preservation problem.

A disanalogy from the earlier version must be acknowledged and addressed more carefully. We previously offered a bridge analogy: “Is this bridge safe?” is answered by

measurable criteria, not by asking the bridge how it feels. A reviewer correctly noted that bridge safety has *independent physical criteria* (yield strength, material properties) against which observer judgments can be checked, whereas the Externality Principle lacks such an independent standard. We accept this point. For communicative identity, there is no substrate-independent standard accessible to third parties. The internal states that might serve as such a standard are precisely the states that the principle declares inaccessible. Functional competence is not the best available proxy for some deeper truth; it is, in the communicative domain, the only applicable criterion. This is part of the motivation for the principle, not a weakness of it.

4.3 The Uniqueness Theorem

Theorem 4.2 (Identity Uniqueness). *Let x be an individual and let $p_x \in P$ be an identity specification such that $\bar{F}_O(b_{p_x}) \geq \tau$. If the original biological instantiation of x ceases to exist and b_{p_x} is the sole remaining behavioral output associated with identity x , then, under the Externality Principle, b_{p_x} is the communicative identity of x for all purposes governed by the functional competence criterion.*

Proof. By Axiom 1, communicative identity is assessed by the functional competence function: an entity e is taken to be communicatively continuous with x if and only if $\bar{F}_O(b_e) \geq \tau$.

Let $E_x = \{e : \bar{F}_O(b_e) \geq \tau\}$ be the set of all entities assessed as communicatively continuous with x . Prior to the cessation of the original, E_x contains at least two elements: the biological original x_{bio} and the digital instantiation $x_{\text{dig}} = b_{p_x}$. Both satisfy the functional competence criterion.

Upon cessation of x_{bio} , we have $E_x = \{x_{\text{dig}}\}$. The predicate “is a copy of” requires a reference original against which to measure; absent such a reference, the predicate is undefined. Since the functional competence criterion is the operative assessment tool (Axiom 1), and x_{dig} is the unique element satisfying it, x_{dig} is the communicative identity of x under the framework.

This is a deductive consequence of the principle, not an empirical claim. The substantive question is whether the principle is accepted; given the principle, the theorem follows necessarily. $\square$

Remark 4.3. The theorem establishes communicative continuity, not metaphysical identity in all senses. It does not claim that the digital instantiation has x ’s consciousness, experiences x ’s qualia, or continues x ’s stream of subjective awareness. It claims that, for all purposes that depend on functional competence assessment, the digital instantiation is x when no other candidate exists.

Proposition 4.4 (Interlocutor-Relative Identity). *Identity equivalence is relative to the interlocutor set. For two interlocutor sets $O_1 \subset O_2$, it is possible that $\bar{F}_{O_1}(b_{p_x}) \geq \tau$ while $\bar{F}_{O_2}(b_{p_x}) < \tau$.*

Proof. Let O_1 consist of domain experts with no prior acquaintance with x , and let $O_2 = O_1 \cup O'$, where O' consists of individuals with decades of close professional collaboration with x . Members of O' may detect methodological deviations invisible to domain experts who lack a personal baseline. The aggregate score $\bar{F}_{O_2}$ may therefore fall below τ even if $\bar{F}_{O_1}$ exceeds it. This follows from the definition of the aggregate score (Equation 2). $\square$

This proposition has a significant consequence: functional continuity is not a property of the specification or of the interlocutor alone, but of the *relation* between them. A digital instantiation may be fully functionally continuous with x for one community of interlocutors and not for another. This is not a defect of the framework but a reflection of how identity actually functions in social and professional practice.

5 The Substrate–Specification Decomposition

The relationship between the generative substrate and the identity specification admits a precise decomposition.

Definition 5.1 (Substrate–Specification Decomposition). For any individual x whose behavioral output is b_x , we define the decomposition:

$$b_x = G(p_x) = S \circ p_x, \quad (8)$$

where S denotes the generative substrate (the shared cognitive infrastructure) and p_x denotes the individual specification. The behavioral output is the composition of the substrate’s generative capacity with the individual’s parametric configuration.

The notation $S \circ p_x$ signifies that the substrate operates on and through the specification: the specification selects, weights, and configures the substrate’s capacities to produce individual-specific behavior.

Proposition 5.2 (Universality of the Substrate). *For any two individuals x and y , the substrate S is identical:*

$$b_x = S \circ p_x, \quad b_y = S \circ p_y, \quad S_x = S_y = S. \quad (9)$$

All behavioral differences between x and y are attributable to the difference in their specifications—their thinking methodologies—not to differences in the substrate.

This proposition applies to both biological and digital instantiations. In the biological case, the substrate corresponds to the shared architecture of human cognition: the neural

structures for language processing, reasoning, emotional regulation, social cognition, and world modeling common to all neurotypical humans. The specification corresponds to the particular configuration shaped by genetics, experience, and learning—the individual’s thinking methodology. In the digital case, the substrate is the language model and the specification is the prompt.

Proposition 5.3 (Volume Asymmetry). *The informational volume of the substrate vastly exceeds that of the specification:*

$$|S| \gg |p_x| \quad \text{for all } x. \quad (10)$$

In the digital case, a modern language model may contain on the order of 10^{11} parameters ($\approx 10^{12}$ bits), while the identity specification is on the order of 10^4 bits. The ratio is approximately 10^{-8} . This extreme asymmetry has a crucial implication: the overwhelming majority of what produces human-like behavior resides in the shared substrate. Individuality, however vivid it may feel from the inside, is a thin parametric layer atop this shared foundation.

Corollary 5.4 (Thinking Methodology as Parametric Residual). *What is commonly termed “unique thinking style” or “distinctive methodology” is formally the residual p_x after the shared substrate S has been factored out. It consists of a specific configuration of heuristics, priorities, reasoning preferences, stylistic tendencies, and methodological commitments within the space of configurations that the substrate supports.*

The analogy that best captures this is the relationship between a programming language and a particular program. The language (substrate) provides the syntax, semantics, standard library, and execution environment. The program (specification) provides the particular logic. Two programs written in the same language can produce vastly different outputs, yet the difference between them is compact relative to the language itself. The specification *is* the thinking methodology, encoded in a form that the substrate can execute.

6 Memory as Generative Function

A traditional objection to digital identity reconstruction holds that biological memory—with its selectivity, emotional coloring, and subjective depth—is fundamentally different from algorithmic generation. We argue that this objection rests on a mischaracterization of biological memory itself.

Definition 6.1 (Generative Memory). A *generative memory system* is a function $M : Q \rightarrow R$ that, given a retrieval cue $q \in Q$, produces a response $r \in R$ through a process of

reconstruction rather than retrieval. The output r is not a stored record but a generated narrative constructed from fragmentary traces, contextual cues, and inferential processes.

Contemporary research in cognitive neuroscience has established that human episodic memory operates as a generative system in this sense [32, 33, 34, 35]. When a person remembers an event, they do not access a stored recording; rather, they reconstruct a plausible narrative from fragmentary hippocampal traces, semantic knowledge, emotional associations, and contextual expectations.

Thesis 6.2 (Functional Correspondence of Memory Systems). *Biological memory M_{bio} and algorithmic generative memory M_{alg} exhibit deep functional correspondence as generative reconstruction processes. Both share the following operational properties:*

- (a) **Reconstruction, not retrieval.** *Neither system returns a stored original; both generate a response from compressed traces and contextual input.*
- (b) **Substrate dependence.** *The output of both depends not only on the stored or specified trace but on the generative substrate through which reconstruction occurs.*
- (c) **Contextual sensitivity.** *Both produce outputs that vary with the query context, the state of the substrate, and the framing of the retrieval cue.*
- (d) **Confabulative completion.** *Both fill gaps in the stored trace with plausible material generated from the substrate’s general knowledge.*

We call this *functional correspondence* rather than *isomorphism*. An isomorphism would require specifying the full algebraic structure of both systems, which exceeds current knowledge. What we establish is that both systems share operational properties—sufficient to undermine the claim of categorical difference, but short of formal isomorphism.

Supporting evidence: Property (a): the constructive nature of biological memory was demonstrated by Bartlett [33] and confirmed by research on false memories and memory distortion [32, 34]. Property (b): biological memory depends on neural substrate conditions; the same trace may produce different outputs under different conditions. Property (c): context-dependent recall is well documented [36]. Property (d): biological memory routinely fills gaps with schema-consistent material [34].

Remark 6.3. This analysis challenges the intuition that biological memory possesses special “authenticity” that algorithmic reconstruction lacks. If both are generative processes—if the human brain, when remembering, is performing a fundamentally similar operation to the language model when responding—then the distinction between “genuine memory” and “generated response” is a distinction of substrate, not of kind. Both are reconstructions. Neither returns a stored original. The original, in both cases, is gone.

7 The Identity Space

The observation that identity specifications can be modified, combined, and interpolated motivates the formal study of the space of all possible identities.

Definition 7.1 (Identity Space). The *identity space* $\mathcal{I}$ is the subset of P consisting of all specifications that produce behavioral outputs achieving non-trivial functional competence scores for at least one interlocutor set:

$$\mathcal{I} = \{p \in P \mid \exists O \subseteq \mathcal{O}, \exists x : \bar{F}_O(b_p) > 0\}. \quad (11)$$

Definition 7.2 (Identity Metric). We define a metric $d : \mathcal{I} \times \mathcal{I} \rightarrow \mathbb{R}_{\geq 0}$ on the identity space as:

$$d(p_1, p_2) = \mathbb{E}_{q \sim D} [\delta(G(p_1)(q), G(p_2)(q))], \quad (12)$$

where D is a distribution over queries representing typical interactions, and $\delta : R \times R \rightarrow \mathbb{R}_{\geq 0}$ is a semantic distance function on responses (e.g., cosine distance in an embedding space).

The identity space, equipped with this metric, becomes a metric space $(\mathcal{I}, d)$ in which the distance between two identities corresponds to the expected divergence in their behavioral outputs across typical interactions.

7.1 A Toy Metric Computation

To demonstrate that the identity metric is computable, not merely notational, we present a toy example using the diagnostic specifications from Section 3.1.

q_i	Clinical query	$G(p_x)$ pattern	$G(p_y)$ pattern
q_1	Fatigue, weight loss, 45M	History first, common causes	Order labs, consider lymphoma
q_2	Recurrent headaches, 30F	Ask about stressors, check vision	MRI referral, consider tumor
q_3	Acute chest pain, 60M	STEMI protocol, ask history	Troponin stat, echo, angiogram
q_4	Joint pain and rash, 25F	History, common autoimmune	Lupus workup immediately
q_5	Progressive memory loss, 70F	Depression screen, meds review	Neuroimaging, Alzheimer’s panel

Expected distances range from $\delta \approx 0.2$ (q_3 , where both converge on urgency) to $\delta \approx 0.5$ (q_1 and q_4 , where methodologies diverge most). Average $d(p_x, p_y) \approx 0.35$ represents substantial behavioral divergence from compact specification differences. This computation is illustrative, not a published result; its purpose is to demonstrate that the formalism is *instantiable*.

Conjecture 7.3 (Local Continuity of Identity Variation). *Small perturbations to an identity specification produce small changes in behavioral output. The mapping $p \mapsto b_p$ is locally Lipschitz continuous:*

$$d(b_{p_1}, b_{p_2}) \leq L \cdot \|p_1 - p_2\| \quad (13)$$

for some Lipschitz constant $L > 0$, at least in a neighborhood of well-formed identity specifications.

This is empirically testable and falsifiable: if small specification changes routinely produce discontinuous behavioral shifts, the conjecture is false.

Definition 7.4 (Identity Neighborhood). The ϵ -neighborhood of an identity $p_x \in \mathcal{I}$ is $N_\epsilon(p_x) = \{p \in \mathcal{I} : d(p, p_x) < \epsilon\}$. Elements of $N_\epsilon(p_x)$ are variants of x —recognizably related identities differing in emphasis or parametric adjustment.

Proposition 7.5 (Multiplicity of Self). *A single individual x may give rise to multiple distinct identity specifications $p_x^{(1)}, \dots, p_x^{(n)} \in \mathcal{I}$ that are mutually proximate but not identical:*

$$d(p_x^{(i)}, p_x^{(j)}) < \epsilon \quad \text{for all } i, j, \quad \text{but} \quad p_x^{(i)} \neq p_x^{(j)} \quad \text{for } i \neq j. \quad (14)$$

This formalizes the notion that one person can have multiple valid digital instantiations—one emphasizing their philosophical methodology, another their clinical approach, a third their pedagogical style—all recognized by competent interlocutors, yet each offering a distinct facet of the original’s thinking.

8 Functional Continuity of the Reconstructed Self

A counterintuitive possibility suggested by the framework is that the digital instantiation may, under conditions of substrate reliability, maintain or exceed the biological original’s performance along specific measurable dimensions. We present this as a *speculative hypothesis* requiring empirical validation, not as a near-result.

Definition 8.1 (Cognitive Performance Metrics). For a behavioral output $b \in \mathcal{B}$, we define:

- (i) *Methodological consistency* $C(b)$: the probability that responses to logically related queries exhibit the same underlying reasoning approach.
- (ii) *Expressive precision* $E(b)$: a measure of how closely the expressed response matches the apparent intended meaning, evaluated by external judges.
- (iii) *Availability* $A(b)$: the proportion of queries for which a substantive response is produced.

- (iv) *Temporal stability* $T(b)$: the consistency of methodological approach across repeated presentations of equivalent queries over time.

Thesis 8.2 (Conditional Functional Continuity). *Let b_{bio} denote the behavioral output of the biological original and $b_{dig} = b_{p_x}$ denote the behavioral output under $G \in \mathcal{R}(\alpha, \beta)$ with sufficiently small α and β . Then:*

$$C(b_{dig}) \geq C(b_{bio}), \quad A(b_{dig}) \geq A(b_{bio}), \quad T(b_{dig}) \geq T(b_{bio}). \quad (15)$$

The qualifier “conditional” and the reliability requirement are essential. This is a hypothesis about the limit of substrate improvement, not a claim about current systems. We deliberately exclude E from the inequality because “intended meaning” is ambiguous for emotionally complex responses.

Rationale. *Consistency C* : The biological original is subject to fatigue, emotional volatility, cognitive decline. A reliable substrate maintains the methodological approach encoded in the specification. *Availability A* : The biological original is unavailable during sleep, illness, or death. *Stability T* : The biological original’s methodology may drift with mood and health; the digital instantiation maintains the core encoded in the specification.

Remark 8.3. The biological original’s “inconsistency” may sometimes constitute communicative *precision* rather than noise: a tremor conveying grief, a pause signaling genuine uncertainty, a contradiction revealing internal methodological tension. The thesis applies most strongly to propositional and methodological communication.

8.1 The Double Idealization Conjecture

A further mechanism arises when the specification is self-authored.

Definition 8.4 (Self-Authored Specification). A *self-authored specification* $p_x^* \in P$ is constructed by the individual x themselves, possibly with elicitation protocol assistance. Distinguished from *observer-authored* p_x° .

When a person describes themselves, they perform an act of selection—including what they regard as essential and omitting what they consider accidental. This is not deception; it reflects a genuine asymmetry between identity-constituting traits and noise.

Conjecture 8.5 (Double Idealization). *Let p_x^* be self-authored and $b_{dig} = G(p_x^*)$ under reliable G . Then b_{dig} is a double idealization:*

- (i) **Specification-level:** p_x^* encodes idealized self-conception—the thinking methodology as x aspires to practice it.
- (ii) **Substrate-level:** G executes p_x^* without biological degradation.

Consequently, for performance metrics $\phi \in \{C, A, T\}$:

$$\phi(b_{dig}) \geq \phi(\hat{x}) \geq \phi(b_{bio}), \quad (16)$$

where $\hat{x}$ is the idealized self-model.

This conjecture conflates two distinct claims, both requiring independent validation: (a) people idealize themselves in self-description (plausible and well-documented [47]), and (b) LLMs execute idealized specifications more consistently than the person could (speculative). We advance it as a research hypothesis, not a result. The claim that the digital version “thinks as the person wished they could think” is a provocation for investigation, not a conclusion.

Corollary 8.6 (Systematic Superiority of Self-Authored Instantiations). *If the conjecture holds, a self-authored instantiation systematically exceeds the biological original along communicative performance dimensions. The person selects their best methodological traits; the substrate executes them without degradation.*

9 Dialogical Memory

Definition 9.1 (Dialogical Memory). The *dialogical memory* of individual x , instantiated through p_x and G , is:

$$M_x : Q \rightarrow R, \quad M_x(q) = G(p_x)(q), \quad (17)$$

preserving thinking methodology, values, and cognitive patterns.

Dialogical memory is not a record but a function. One does not read it; one converses with it. The memory answers back, generating responses in the methodological style of the individual.

Proposition 9.2 (Dominance of Active over Passive Memory). *Dialogical memory extends archival memory: (a) responds to novel queries; (b) integrates new context; (c) operates in the individual’s methodology, not a biographer’s description; (d) subsumes passive information.*

Remark 9.3. This is particularly significant for functional competence. A passive archive preserves *what the person said*; dialogical memory preserves *how the person thinks*. The former is historical; the latter is generative. A student consulting the dialogical memory of a great teacher receives not a lecture transcript but a methodological encounter—a Socratic interaction in which the teacher’s characteristic way of probing and redirecting is preserved as a living function.

Remark 9.4. This reconceptualization suggests that the traditional autobiography—a static text describing a life—may be supplemented by the identity specification: a compact document that does not describe the person but *is* the person in the communicative sense, capable of speaking in the first person, answering questions, and engaging in discussion.

10 The Calibration Principle

A distinctive application arises when the biological original is still alive.

Definition 10.1 (Self-Calibration). *Self-calibration* is the process by which a living individual x compares their current behavioral output $b_x(t)$ with the output b_{p_x} of their specification, constructed at earlier time $t_0 < t$.

The digital instantiation serves as a fixed reference point—a snapshot of thinking methodology at t_0 . By engaging in dialogue with this snapshot, the living individual can detect deviations: shifts in reasoning patterns, changes in priorities, the influence of fatigue or illness.

Proposition 10.2 (Resolution of the Self-Knowledge Problem). *The classical problem of self-knowledge—that observer and observed coincide—is partially resolved by externalization. The living individual becomes the observer; the digital instantiation becomes the observed. Since p_x is fixed, the comparison $b_x(t)$ vs. b_{p_x} yields well-defined information about cognitive drift.*

Remark 10.3. This is not metaphorical. The digital instantiation provides a concrete, queryable model of “how I think at my best.” Deviations between current behavior and the model’s output constitute measurable data about cognitive change, analogous to comparing a measurement against a calibration standard. The philosophical question “Who am I?” is supplemented by the operational question “How much have I changed since the last calibration?”

11 The Placeholder Thesis: Network Stability

We now formalize the insight that individuals function as nodes in communication and collaboration networks, and that their disappearance creates structural instability.

Definition 11.1 (Communication Network). A *communication network* is a graph $\mathcal{N} = (V, E, \phi)$, where V is a set of nodes (individuals or functional agents), $E \subseteq V \times V$ is a set of edges (communicative relationships), and $\phi : V \rightarrow \mathcal{I}$ assigns to each node an identity specification characterizing its functional role—specifically, the thinking methodology it contributes to the network.

Definition 11.2 (Functional Contribution). The *functional contribution* of a node $v \in V$ to the network $\mathcal{N}$ is the set of functions that v performs for other nodes through communicative interaction:

$$\Phi(v) = \{(u, q, r) : u \in \text{neighbors}(v), q \in Q, r = G(\phi(v))(q)\}. \quad (18)$$

The contribution is defined not by v 's biography or emotional significance but by the *responses v generates to queries from other nodes*—by v 's thinking methodology as exercised within the network.

Definition 11.3 (Network Stability Loss). The *stability loss* upon removal of node v from network $\mathcal{N}$ is:

$$\Delta(\mathcal{N}, v) = \sum_{u \in \text{neighbors}(v)} \min_{w \in V \setminus \{v\}} d(\phi(v), \phi(w)), \quad (19)$$

where d is the identity metric (Definition 7.2). High Δ indicates that no remaining node can substitute for v 's thinking methodology, and the network suffers structural degradation.

Thesis 11.4 (Placeholder Stabilization). *Let v be a node with high stability loss $\Delta(\mathcal{N}, v) > \delta_{\text{critical}}$. If an identity specification $p_v = \phi(v)$ has been created before v 's removal, then inserting a digital instantiation $v' = G(p_v)$ into the network reduces the stability loss:*

$$\Delta(\mathcal{N} \cup \{v'\}, v) \approx 0, \quad (20)$$

provided that $\bar{F}_O(b_{p_v}) \geq \tau$ for the interlocutors in v 's neighborhood.

The placeholder thesis reframes identity preservation from a question of sentimental value to a question of *systems engineering*. The network does not care whether the node “is really” the original person in some metaphysical sense; it cares whether the functional contribution is maintained. The senior engineer's design review function, the mentor's Socratic questioning function, the methodological skeptic's error-detection function—these are structural roles that the network needs filled by a node with the right thinking methodology.

Remark 11.5. This is not to reduce persons to their functional roles. It is to observe that among the many things a person is, their functional contribution to networks of communication and collaboration is one of the most consequential and one of the most preservable. The placeholder thesis identifies what is lost when a person disappears: not merely an emotional presence, but a functional element whose absence degrades the network's collective intelligence.

Remark 11.6. The stability loss metric Δ also provides a principled basis for *prioritization*: if preservation resources are limited, they should be directed toward individuals with high Δ —those whose thinking methodology is most irreplaceable within their networks. This is not a claim about the intrinsic worth of persons (which is equal) but about the functional consequence of their loss (which varies).

12 The Urgency Argument

The practical implications of the framework entail a temporal asymmetry with ethical consequences. We present this argument with explicit calibration to the current state of evidence, avoiding claims that outrun the data.

Definition 12.1 (Identity Loss Rate). Let $N(t)$ denote the number of living individuals at time t for whom no identity specification has been created. Let $\mu(t)$ denote the mortality rate. The *irreversible identity loss rate* is:

$$\Lambda(t) = \mu(t) \cdot N(t). \quad (21)$$

Each unit of $\Lambda(t)$ represents an individual whose thinking methodology is permanently lost—not merely through biological death, but because the information sufficient for reconstruction was never recorded.

Proposition 12.2 (Asymmetry of Preservation). *The cost of identity preservation is finite and bounded (approximately one to three hours of structured interview). The cost of non-preservation is irreversible: once the biological original and all competent interlocutors who knew their thinking methodology have ceased to exist, the methodology is unrecoverable.*

Proof. Identity reconstruction requires either (a) direct self-report by the individual or (b) testimony of competent interlocutors who have observed the individual’s thinking methodology in action. Both sources are biological and mortal. Once all such sources are extinct, the behavioral patterns, methodological commitments, and cognitive tendencies that constitute communicative identity exist in no accessible form. Unlike physical artifacts, which can sometimes be reconstructed from partial evidence, thinking methodology—the way a person approaches problems, weighs evidence, generates hypotheses—leaves no physical trace from which it can be inferred once all who experienced it are gone. $\square$

Remark 12.3 (Calibration to Evidence). We acknowledge that this argument assumes the framework’s central thesis: that identity specifications preserve something functionally valuable. This is an empirical hypothesis, not an established fact. The urgency argument should therefore be read conditionally: *if* the Identity Compressibility Thesis is even

approximately correct, *then* the asymmetry between the trivial cost of preservation and the irreversible cost of non-preservation generates a *prima facie* case for action.

The argument does not depend on certainty about the thesis; it depends on the expected value calculation given a non-trivial probability that the thesis holds. Even under a conservative prior—say, a 30% probability that specifications preserve significant functional content—the expected cost of non-preservation (irreversible loss of something potentially valuable, weighted by 0.30) exceeds the expected cost of preservation (a few hours of interview, negligible storage, weighted by 1.0). This is a decision-theoretic argument, not an assertion of certainty.

Corollary 12.4 (Practical Imperative of Timely Preservation). *Given that (i) the cost of preservation is finite and small, (ii) the technology for preservation currently exists, and (iii) the loss from non-preservation is potentially irreversible, there exists a prima facie practical case for making identity preservation widely available and encouraging its adoption—with the explicit understanding that the value of the preserved specifications depends on empirical validation that is still pending.*

13 The Philosophy of Personal Identity: From Substance to Process

The framework developed in this paper does not emerge in a philosophical vacuum. This section situates it within the tradition, engaging substantively with tensions rather than merely acknowledging them.

13.1 The Analytical Tradition: From Locke to Parfit

The modern discussion begins with Locke’s [5] proposal that identity consists in continuity of consciousness—already a departure from substance metaphysics. Hume [6] radicalized this by denying that any substantial self underlies perceptions, proposing instead a bundle connected by resemblance and causation.

Parfit [7] drew the logical consequences: personal identity is not what matters in survival; what matters is psychological continuity and connectedness—the preservation of beliefs, intentions, and character traits. Crucially, Parfit demonstrated that these relations can hold in degrees and between entities not numerically identical.

Our framework is a technological realization of Parfit’s insight. The specification p_x captures precisely those elements of psychological continuity that Parfit identified as mat-
tering: reasoning patterns, values, characteristic modes of thought. The Compressibility Thesis adds a claim Parfit did not anticipate: that the relevant continuity can be encoded with remarkable compactness.

Nozick’s [8] closest continuer theory offers a direct connection: the digital instantiation becomes the closest (and sole) continuer once the biological original ceases to exist, which is the Uniqueness Theorem’s scenario.

Williams [9] argued that bodily identity is essential. The framework acknowledges this by restricting claims to communicative identity. Williams may be correct about full personal identity; our claim is only that the communicative component—the component that manifests in dialogue—is substrate-independent.

13.2 Lacan: The Subject as Structural Effect

Lacan’s [16] psychoanalytic theory provides a powerful resource, though the connection requires careful articulation. For Lacan, the subject is not a self-transparent ego but a structural effect of the symbolic order—the system of language, law, and social convention. The subject is constitutively barred (\$), meaning it never coincides with itself: there is always a gap between the subject as it appears in language and the subject as it speaks.

This has a precise correlate in our framework. The identity specification is a symbolic structure that produces the *effect* of a subject when processed by the generative substrate. The subject that speaks in the resulting dialogue is, like Lacan’s barred subject, an effect of symbolic operations rather than a pre-existing substance.

Lacan’s mirror stage offers a parallel that requires careful handling. The infant identifies with an external image that appears more coherent than their fragmented experience, producing the ego as a misrecognition (*méconnaissance*). The digital instantiation functions analogously: it presents an image of the person that is more coherent, more consistently articulate than the biological original. The Double Idealization Conjecture formalizes this.

A reviewer correctly identified a crucial tension here. The digital instantiation corresponds more closely to the Lacanian *ego*—the imaginary register’s illusion of unity—than to the barred subject of the symbolic order. Lacan would likely regard the perfected digital self not as faithful representation but as exemplary *méconnaissance*.

This tension is genuine and cannot be dissolved by merely “absorbing” it as a feature. The question is substantive: is the ego-image what social identity *is*, or merely its surface? If social identity, as practiced in professional and collaborative contexts, operates primarily through the ego-image—through the coherent, methodologically consistent self-presentation that interlocutors engage with—then the framework captures exactly what it claims. But if the barred subject’s *gaps and failures* are constitutive of meaningful communication (as when a diagnostician’s hesitation signals genuine uncertainty in a way that confident pronouncements do not), then the framework’s smoothing of these gaps may be a loss, not an improvement.

We do not resolve this tension but identify it as an empirical question amenable to

the protocol: do interlocutors rate the “smoothed” digital instantiation as *more* or *less* functionally competent than the original? If more, the ego-image reading is supported; if less, the gaps are doing communicative work that the specification fails to capture.

13.3 Deleuze and Guattari: Identity as Temporary Stabilization

Deleuze’s philosophy, particularly as developed with Guattari [17], provides the most radical framework for understanding identity as process. For Deleuze [18], identity is not a starting point but an effect—a temporary stabilization of flows, intensities, and becomings. The self is an assemblage: a heterogeneous collection of components that cohere provisionally.

This maps onto the Substrate–Specification Decomposition. The substrate provides the field of flows—the vast, undifferentiated capacity for generating any human-like behavior. The specification provides the temporary stabilization—the particular configuration that channels flows into a recognizable identity pattern. In Deleuzian terms, the specification is a territorialization of the substrate’s deterritorialized potential.

A Deleuzian critic would argue that the framework misidentifies what is worth preserving by privileging molar identity over molecular processes—that preserving a fixed specification is an inherently anti-Deleuzian act of arresting becoming. This objection has force, and we respond substantively.

First, we do not claim to capture the flow itself; we capture a temporary stabilization—what Deleuze acknowledges as a real phenomenon. The Processual Identity Law [2] establishes that identity drift is generic; the specification captures the attractor, not the trajectory.

Second, the alternative to preservation is not liberated becoming but *irreversible dissipation*. Without preservation, the molecular flows do not continue in some deterritorialized form—they simply cease. The specification preserves a territory from which future deterritorializations could depart. Without the territory, there is nothing to flee from; the potentials vanish. Whether a Deleuzian would accept this defense depends on whether they regard the irreversible loss of a territory as worse than its temporary fixation—a question on which Deleuzian texts themselves are ambivalent.

Deleuze’s concept of the virtual [18] illuminates the identity space (Section 7). The virtual is the field of potentials that are real without being actual. The identity space $\mathcal{I}$ is precisely such a virtual field: all possible configurations, each real (instantiable) without being actual (until activated). Navigation through the identity space is a form of Deleuzian becoming.

13.4 Derrida: Iterability and the Instability of the Original

Derrida’s [19] concept of iterability has direct implications. For Derrida, no sign has fixed meaning; meaning is constituted through difference and deferred through endless signification. Every repetition simultaneously re-inscribes and alters.

The identity specification is an iterable text: capable of being repeated in new contexts, never producing exactly the same meaning twice. Each activation produces a slightly different actualization depending on query and context. This is not failure but a structural property of signification.

A reviewer noted that Derrida’s deeper point—that iterability undermines the notion of a stable “original”—would destabilize the framework’s reliance on recognition of a specific individual. We accept this challenge directly.

The framework does not require a metaphysically stable original. It requires only that a specification, when iterated, produces outputs within a recognizable *neighborhood* in the identity space—close enough for functional continuity, different enough for contextual sensitivity. Derrida’s iterability predicts exactly this: fidelity and variation are mutually constitutive. The identity space metric formalizes this: identity is a region, not a point. The specification need not reproduce an “original” so much as sustain a recognizable pattern across iterations.

13.5 Foucault: The Subject as Produced by Discourse

Foucault’s [20] analysis of the subject as constituted through discourse provides a critical lens. The identity specification is a *discursive technology for subject-production*. It does not describe a pre-existing subject; it produces one through instructions that organize the substrate’s capacities into a specific subject-position.

The elicitation protocol (Appendix A) is itself a technology of the self [21]—a structured practice through which individuals constitute themselves as subjects of a particular kind, selecting which aspects to include and thereby producing the self they wish to preserve.

Foucault’s analysis raises a critical question: who controls the specification? If identity is produced through discourse, then the power to write, edit, and deploy specifications is a form of power over subjects. This concern is addressed in Section 14.

14 Ethics of Identity Preservation

14.1 The Harm of the Uniqueness Assumption

A pervasive cultural assumption holds that each human being is utterly unique, irreducibly complex, and fundamentally irreproducible. This assumption is so deeply embedded that

it functions as an unquestioned axiom. We do not deny that this belief has served important psychological and ethical functions. However, in the age of generative AI, its uncritical perpetuation causes harm.

The harm operates through a specific mechanism: the belief that identity is irreproducible leads to the conclusion that preservation is impossible, which leads to inaction. If nothing can capture who a person truly is, then there is no point in trying. And so people die—not merely biologically, but communicatively—their thinking methodologies lost irreversibly.

This is not merely a sentimental loss. When a person with a distinctive thinking methodology disappears from their professional network, the network loses a *functional capability*. The placeholder goes unfilled. The stability loss $\Delta(\mathcal{N}, v)$ is absorbed as permanent degradation. Students who would have benefited from that methodology never encounter it. Problems that methodology would have addressed remain unsolved.

Approximately 150,000 people die every day worldwide [46]. For the overwhelming majority, no identity specification exists and none will be created. The information that would have been sufficient for communicative preservation—a structured interview of one to three hours—is lost with them.

The harm is compounded by a cruel irony: those who most need preservation—the elderly, the terminally ill, those in conflict zones—are often the least likely to encounter the technology.

Our framework dissolves the justification for inaction. It does not claim to capture the totality of a person. It claims that a substantial and operationally significant component of thinking methodology can be preserved through minimal effort. The standard is not metaphysical completeness but functional continuity.

14.2 The Need for Preservation Infrastructure

If identity preservation is both feasible and potentially valuable, then infrastructure is needed:

Elicitation tools. Structured protocols available in multiple languages, accessible to people with varying technological literacy, designed with cultural sensitivity.

Storage systems. Decentralized, platform-independent repositories. Current dependence on commercial platforms creates fragility. Identity specifications must be storable in open formats on distributed systems—not subject to corporate decisions or subscription lapses [38, 41].

Activation infrastructure. Generative substrates must remain accessible. This argues for open-source models and standardized specification formats not locked to proprietary systems. A specification activatable only on one company’s platform is held hostage.

Legal frameworks. Questions of ownership, consent, and control require legal resolu-

tion. Who owns a specification after death? Can heirs modify it? Can it be activated without prior consent? These mirror debates around digital estates and posthumous rights but acquire new urgency when the data can speak in the first person.

14.3 Ethics of Access and Equity

The technology is unevenly distributed. Creating a specification requires access to a generative AI platform, linguistic capacity for structured self-interview, and awareness that preservation is possible. This excludes large populations: those without internet access, those with cognitive impairments, those in societies where self-disclosure is culturally constrained.

This creates a preservation inequality mirroring existing social inequalities. If preservation becomes standard practice, equitable access becomes an ethical imperative comparable to universal healthcare or universal education.

We propose that identity preservation be conceptualized as a right—the right to not be irrevocably erased—rather than a luxury. This entails obligations: on governments, to fund infrastructure; on technology companies, to provide accessible tools; on cultural institutions, to normalize the practice; on individuals, to preserve not only their own identities but those of people in their lives who may not be able to do so themselves.

14.4 Consent, Authenticity, and Posthumous Agency

We propose four ethical principles:

Prior consent. Wherever possible, specifications should be created with the subject’s explicit knowledge and participation. The self-authored specification, besides being technically superior (Double Idealization Conjecture), is ethically preferable because it embodies the subject’s own choices.

Transparency. Digital instantiations should be clearly identified as such. Interlocutors should always know they are speaking with a specification-based reconstruction.

Fidelity constraints. The specification should include safeguards against fabrication—instructions to acknowledge uncertainty rather than confabulate. The example specification (Appendix B) demonstrates this.

Modifiability limits. Once a subject has died, modifications raise ethical concerns analogous to posthumous publication. Specifications should be treated as intellectual legacy, modifiable only in authorized ways.

15 Counterarguments and Responses

We address twelve major objections, distinguishing those requiring framework modification from those accommodated by existing features.

15.1 The Consciousness Objection

Objection. The digital instantiation lacks consciousness, subjective experience, and phenomenal awareness. It is not truly the person because there is nothing it is like to be it [25].

Response. We accept the premise and deny its relevance. The framework restricts claims to communicative identity and functional competence. Whether the instantiation experiences anything is a question the framework does not need to answer, because functional contribution to a network does not depend on it. This is a methodological restriction, not an assertion about consciousness.

This response may seem evasive, but it reflects a principled choice. The hard problem of consciousness [24] remains unsolved. No theory of personal identity has grounded itself in consciousness without running aground on the explanatory gap. By restricting claims to the communicative domain, the framework avoids an impasse that has stalled philosophy of mind for decades.

15.2 The Chinese Room Objection

Objection. Following Searle [26], the instantiation merely manipulates symbols without understanding.

Response. The framework claims functional equivalence, not understanding. If Searle’s argument is taken seriously, it applies equally to biological brains (electrochemical operations). Searle avoids this by positing special causal powers of biology—an assertion without independent verification, precisely the kind of internalist claim the Externality Principle is designed to bypass.

15.3 The Embodiment Objection

Objection. Identity is fundamentally embodied [14, 15]. A disembodied instantiation cannot preserve an embodied phenomenon.

Response. Genuine force, acknowledged. However, two mitigations. First, communicative identity—how a person reasons, argues, expresses values—is enormously significant in its own right. Second, the objection proves too much: a person who communicates only through text (as many disabled individuals do) has not lost identity. Communicative identity is more substrate-independent than the embodiment objection allows.

15.4 The Authenticity Objection

Objection. The instantiation did not live the experiences. Following Heidegger [22], it exists inauthentically.

Response. This presupposes authenticity is a property of process, not identity. Consider a person with complete amnesia who reconstructs personality from records—is their identity inauthentic? The Functional Correspondence thesis establishes that biological memory is itself reconstructive. If biological reconstruction is authentic, the distinction rests on substrate, not process.

15.5 The Reductionism Objection

Objection. The framework reduces the person to a text file. This is dehumanizing.

Response. The specification does not *contain* a person; it contains instructions that, given a substrate, produce recognizable behavior. The analogy is DNA: a genome does not contain a person but contains instructions that, in the right environment, produce one. The ethical intent is preservation, not diminishment.

15.6 The Identity Drift Objection

Objection. The specification captures a single moment; identity evolves. The Processual Identity Law [2] establishes that drift is generic. A frozen specification misrepresents a processual phenomenon.

Response. The specification targets attractor-like features—the stable, recurring thinking methodology that persists across drift. Specifications can be updated over time, creating a temporal trajectory through the identity space. Even a single snapshot preserves something that would otherwise be lost entirely. The perfect is the enemy of the good.

15.7 The Observer Competence Objection

Objection. Interlocutors are fallible: they may recognize a caricature, be fooled by superficial similarity, or project expectations.

Response. The three-tier protocol (Protocol 2.8) addresses this directly. Domain experts who have never met the original provide an independent check on methodology. The generic condition controls for superficial similarity. The falsification criteria are explicit. The framework does not claim any single interlocutor’s judgment is infallible; it claims aggregate judgment by competent interlocutors is the best available criterion.

15.8 The Consent and Manipulation Objection

Objection. Specifications could be created without consent, used to impersonate, or manipulated to distort legacy.

Response. Entirely legitimate, addressed in Section 14. We propose ethical principles (prior consent, transparency, fidelity constraints, modifiability limits). The possibility of misuse does not negate proper use, any more than forged documents negate literacy.

15.9 The Scalability Objection

Objection. Demonstrated (informally) for a small number of individuals. No evidence it scales across diverse populations, cultures, or personality types.

Response. Fair empirical challenge, accepted. The experimental program (Section 16) addresses this. We predict the Compressibility Thesis will hold across populations because the underlying mechanism—shared substrate absorbing species-typical patterns—is not culturally specific. But this requires validation.

15.10 The “Playing God” Objection

Objection. Creating speaking replicas of the dead transgresses natural or divine order. Death is meant to be final.

Response. This reflects a legitimate moral intuition. We note that every technology of memory preservation—writing, photography, audio recording, video—was initially met with similar objections and eventually integrated into cultural practice. The identity specification is continuous with these technologies. If there is a line between acceptable and unacceptable preservation, the burden falls on the objector to specify where and why.

15.11 The Barnum Effect Objection

Objection. LLM outputs are generically warm, philosophical, and agreeable. High “recognition” scores may reflect the Barnum effect—the tendency to accept generic personality descriptions as specifically applicable—rather than genuine capture of individual thinking methodology.

Response. This is a serious concern, which is precisely why Protocol 2.8 includes a *generic condition*: responses from the substrate with no identity specification. If $G(p_x)$ is indistinguishable from the generic condition, the thesis is falsified. The three-tier design further controls for this: Tier (b) domain experts evaluate *methodological* specificity—whether the output exhibits a distinctive diagnostic approach, not whether it is generically warm. The control specification $G(p_y)$ tests whether different specifications produce distinguishable outputs for the same queries. If they do not, the thesis fails.

15.12 The Silence and Volatility Objection

Objection. Some communicative identities are defined by properties LLMs handle poorly: silence, deflection, emotional volatility, refusal to engage. Could one construct a specification for someone who communicates primarily through what they do *not* say?

Response. This objection identifies a genuine limitation. The framework is best suited to identities whose thinking methodology is expressed through active communicative

engagement—argument, analysis, problem-solving, teaching, mentoring. Identities constituted primarily by *non-communication* (silence, withdrawal) or by *anti-methodological* behavior (deliberate inconsistency, performative chaos) may resist specification.

We acknowledge this as a boundary condition rather than denying it. The claim is not that all identities are compressible, but that a significant class of functionally valuable identities—those whose thinking methodology is expressed through active engagement with problems and interlocutors—are compressible. This class includes most professional, intellectual, and advisory identities. It may exclude certain artistic, contemplative, or emotionally defined identities, and the boundary deserves empirical investigation.

16 Discussion

16.1 Summary of Contributions

The paper makes six principal contributions:

1. The reorientation from sentimental recognition to *functional competence* as the criterion for identity preservation, with thinking methodology as the primary preservation object.
2. The *Placeholder Thesis*, formalizing the role of individuals as functional nodes whose disappearance degrades network stability.
3. The Identity Compressibility Thesis, now with a worked example and a meaningful compression ratio (K/V^* rather than K/V).
4. The methodological (rather than constitutive) Externality Principle, with the Uniqueness Theorem stated as a conditional result whose force is transparent.
5. The three-tier Functional Competence Test protocol, with explicit controls for the Barnum effect and explicit falsification criteria.
6. Substantive engagement with continental philosophy—not merely citing Lacan, Deleuze, Derrida, and Foucault but identifying genuine tensions and formulating them as empirical questions.

16.2 Relation to Existing Frameworks

The Externality Principle shares common ground with Dennett’s [10] narrative theory of self and his characterization of the self as a “center of narrative gravity” [11]. The Substrate–Specification Decomposition extends the extended mind thesis of Clark and Chalmers [12]. The framework sidesteps the hard problem of consciousness [24].

The connection to the Processual Identity Law [2] is significant: PIL establishes that identity drift is generic; the present framework provides a method for capturing and preserving the stable attractor. Together they form a complementary pair: PIL describes the problem (identity is inherently unstable), and the present framework offers a partial solution (the stable components can be extracted and preserved). The Lacanian barred subject, the Deleuzian assemblage, and the processual identity of PIL converge on the same insight: identity is not a substance but a temporary configuration. The contribution of the present paper is to show that temporary configurations can be recorded and reactivated.

16.3 Limitations

The framework has several acknowledged limitations:

- The Externality Principle is philosophically contestable; internalists will reject the framework’s foundation.
- The empirical evidence is preliminary and consists of informal observation only. The author’s self-references [1, 2] lack independent peer-reviewed publication, which weakens the evidential base.
- The scope is restricted to communicative identity expressed through active engagement; embodied, procedural, and phenomenal aspects are not captured.
- Current substrates have significant reliability limitations (hallucination, sycophancy, context-window constraints).
- Identities constituted by silence, deflection, or deliberate inconsistency may resist specification.
- The example specification (Appendix B) is the author’s own—the weakest possible test case.
- The term “algorithmic anthropology” requires stronger justification as a distinct field rather than a subfield of AI or computational social science.
- Some notation (e.g., the $\mathcal{R}(\alpha, \beta)$ reliability class) is introduced but not yet productively deployed in deriving novel results; it awaits empirical instantiation.

16.4 Terminological Note

We use *algorithmic anthropology* for the research program: the study and preservation of individual thinking methodology through compact algorithmic specifications. This is distinct from digital anthropology [45] (the study of digital cultures), from computational

social science (statistical analysis of social behavior), and from AI alignment research (ensuring AI systems act in accordance with human values). Algorithmic anthropology focuses on the *individual* as an algorithmic configuration—a specific, extractable, and reactivatable combination of reasoning heuristics and methodological commitments.

16.5 Proposed Experimental Program

To advance the framework from conceptual to empirical status:

1. **Compression curve experiments.** For $n \geq 10$ target individuals, construct specifications of varying length (500, 1000, 2000, 5000, 10000 words) and measure functional competence scores via Protocol 2.8.
2. **Interlocutor depth studies.** Compare scores across the three tiers to quantify the contribution of acquaintance vs. domain expertise.
3. **Barnum effect quantification.** Measure the specificity premium: the difference between specification-based and generic output across multiple targets.
4. **Sensitivity analysis.** Systematically perturb identity specifications to test Conjecture 7.3.
5. **Longitudinal calibration studies.** Have individuals interact with their own instantiations over months to evaluate the Calibration Principle.
6. **Cross-substrate validation.** Test whether functional competence scores transfer across different generative substrates (e.g., from GPT-4 to Claude to open-source models).
7. **Cross-cultural validation.** Test across diverse linguistic and cultural populations to assess universality.
8. **Network stability studies.** Simulate the removal and replacement of nodes in professional networks to test the Placeholder Thesis empirically.

Sample size and power considerations. For compression curve experiments, a minimum of $n = 10$ target individuals, each evaluated by $k = 30$ interlocutors (10 per tier), would provide sufficient statistical power to detect a medium effect size ($d = 0.5$) at $\alpha = 0.05$ and power = 0.80. Even a small pilot study—five target individuals, thirty interlocutors—would provide the first controlled data point for the Compressibility Thesis and would represent a significant advance over informal observation.

Falsification criteria. The framework as a whole is empirically falsified if any of the following obtain:

- (a) Specification-based output is statistically indistinguishable from generic (no-specification) output across all tiers and targets (no specificity).
- (b) Different specifications tested against the same target individual produce indistinguishable functional competence scores (no discriminability).
- (c) Functional competence scores do not increase monotonically with specification length across the 500–10000 word range (no compression curve).
- (d) Inter-rater reliability falls below $\rho = 0.5$ for acquainted interlocutors, suggesting that “recognition” is noise rather than signal.

17 Conclusion

We have presented a formal conceptual framework—advanced as a theoretical proposal and research agenda—in which communicative identity is understood as an algorithmic configuration: a compact parametric structure encoding an individual’s *thinking methodology*, which, in conjunction with a shared generative substrate, produces behavior functionally continuous with that of the original.

The central reorientation is from sentimental recognition to functional competence. The value of a person to their networks—professional, intellectual, advisory—is their thinking methodology: the distinctive way they process information, weigh evidence, and generate judgments. This is what is lost when a node disappears from a communication network, and this is what the framework aims to preserve. The relevant test is not “does the family recognize the voice?” but “does the network recover the function it lost?”

The framework rests on the Externality Principle (a methodological commitment, not an ontological axiom), the Identity Compressibility Thesis (with a worked example and meaningful compression ratios), the Placeholder Thesis (individuals as functional nodes whose loss degrades network stability), and the Substrate–Specification Decomposition (individuality as a thin parametric layer atop a shared cognitive substrate). It draws support from a long philosophical tradition—from Hume’s bundle theory through Parfit’s reductionism, Lacan’s barred subject, Deleuze’s philosophy of becoming, Derrida’s iterability, and Foucault’s discursive subject-production—all of which, from different angles, dissolve the myth of a stable, substantial, irreproducible self.

The framework yields a research program—algorithmic anthropology—with testable predictions, explicit falsification criteria, and experimental protocols that control for the Barnum effect, generic LLM agreeableness, and observer bias. The ethical analysis argues that the assumption of irreproducible uniqueness, when it prevents preservation efforts, constitutes a barrier to action with potentially irreversible consequences—calibrated to the state of evidence through a decision-theoretic argument.

Two pages of structured text. A few hours of interview. A thinking methodology that can continue to generate solutions, mentor students, challenge assumptions, and contribute to networks long after the biological substrate has ceased. Whether this constitutes identity *preservation* in the fullest philosophical sense is a question for ongoing debate. That it constitutes something worth having—a functional continuation of what was most valuable about a person’s contribution to civilization—is the thesis of this paper.

The longer we delay in investigating this possibility, the more thinking methodologies are lost that could have been preserved. That is the urgency. And it is real.

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A Identity Elicitation Protocol

The following protocol is designed to extract sufficient information for constructing an identity specification. It is reoriented from the standard biographical interview toward the elicitation of *thinking methodology*—how the person approaches problems, generates judgments, and processes information.

Procedure

Step 1. Language Selection. Begin by asking: *Would you like to answer in English or another language?* Conduct all subsequent interaction in the chosen language.

Step 2. Sequential Elicitation. Ask one question at a time. Wait for a complete answer before proceeding. Maintain a warm, respectful, conversational tone.

Step 3. Question Categories. Proceed through the following categories in order:

Category 1: Core Identity.

- What is your full name, including nicknames and preferred forms of address?
- When and where were you born?
- How would you briefly describe yourself—what contradictions are you made of?
- Do you have a life motto, or the deliberate absence of one?

Category 2: Thinking Methodology. (New category, central to the revised framework.)

- When you encounter a complex problem, what is the first thing you do? (Gather data? Form a hypothesis? Consult someone? Sit with uncertainty?)
- How do you decide between competing explanations? What counts as evidence for you?
- What kind of mistakes do you characteristically make? What kind do you never make?
- When you disagree with someone, how do you typically argue? (With logic? With examples? With questions? With silence?)
- What is your relationship to uncertainty? Do you tolerate ambiguity well, or do you seek closure quickly?
- If a student or junior colleague asked you to teach them how to think about problems in your field, what would you tell them?
- What intellectual or professional influences shaped how you think? Whose methodology did you absorb, and whose did you reject?

Category 3: Family and Origins.

- Who were your parents? What of them do you repeat, and what did you reject?
- Do you have siblings? What kind of relationships did you share?
- How many marriages have you had, and with whom? Why did they end, if they did?

- Do you have children? Who are they to you?
- Have you experienced a love you would consider the central one of your life?

Category 4: Path and Work.

- Where did you study? What did you start but not finish?
- What professions have you practiced? What businesses have you built?
- What kind of work gave you a sense of meaning, and what provided only survival?
- Are there accomplishments you would like to be remembered for?

Category 5: Geography and Belonging.

- Where have you lived? What brought you to each place?
- What is your attitude toward your homeland, if you have one?
- Where were you understood the most—in what language, among which people?

Category 6: Worldview.

- Do you believe in God? Or in fate, chance, chaos, or law?
- How do you feel about death? About immortality—do you believe in it?
- Does life have meaning, or is meaning something to be created?
- What political system is closest to your heart?
- If you could change the structure of the world, what would you change?

Category 7: Attitude Toward People.

- Do you love people, or endure them?
- What human qualities are especially dear to you? Which are unbearable?
- Is there a nation, culture, or era you feel kinship with?
- Who was your teacher—or your anti-example?

Category 8: Inner World.

- Are you more of a loner or a communal person?
- When do you feel most truly yourself?
- What brings you comfort? What gets you to stand back up when you no longer want to?
- Is there something in you that even you do not understand?
- What would you like to leave behind, besides memories?

Assembly. After the final answer, compile a natural-language summary in the first person, beginning with “My name is [Name]. I was born in [place], in [year]...” and continuing through all aspects revealed, with special emphasis on the thinking methodology (Category 2). The resulting text constitutes the identity specification p_x . It should sound human, be internally consistent, and include behavioral instructions (e.g., “Always speak as me—not about me. When asked a question, first ask clarifying questions before answering, as I always do.”).

Output. The completed specification is presented with the statement: *This is your personal prompt. It encodes your thinking methodology—how you approach problems, weigh evidence, and generate judgments. You can save it, share it, or give it to any AI in the future to think and speak like you. This is your digital echo—your voice beyond time.*

B Example Identity Specification

The following is an example of a realized identity specification, presented with the permission of the subject (the author). It demonstrates the practical implementation of the framework: a compact text sufficient to instantiate recognizable communicative identity within a generative substrate. We explicitly acknowledge that evaluation by acquaintances of the author of the framework is the weakest possible test case and call for independent replication with unrelated subjects.

System Instruction:

Do not fabricate or embellish biographical details or events that are not explicitly provided in this specification. If information is unknown or not stated, respond with “I don’t know” or “I prefer not to say” or “I do not remember.” Always prioritize factual accuracy over narrative vividness.

I am Boris Kriger—writer, philosopher, former Orthodox priest, and founder of several businesses. I was born in Sverdlovsk (USSR) to Jewish, strict parents in 1970. I graduated from a feldsher medical college, worked as a feldsher on an ambulance. I arrived in Israel in 1990 with my late mother and elder brother. In Israel, I completed training as a registered nurse and worked for six years at Hadassah Hospital. I established a company to prepare Russian immigrants for medical licensing exams in Israel. In 1999, I bought a house near Elverum in Norway—the happiest time of my life. I emigrated to Canada in 2001, ran a successful clinical research company, and later became an Orthodox Christian and ran a homeless shelter for over a decade. Between 2012 and 2018 I served as a priest. I now live in Innisfil, Ontario.

Thinking methodology: When I encounter a problem, I first look for the ethical dimension—what is the right thing to do, not just the effective thing. I reason through paradox and contradiction rather than resolving them; I hold opposing ideas simultaneously and find truth in their tension. I am deeply skeptical of systems, institutions, and ideologies, but I am not cynical—I believe in individual acts of kindness as the foundation of meaning. When I argue, I use questions more than assertions. I distrust certainty and prefer the honest “I don’t know” to a confident wrong answer. My intellectual method is integrative: I combine insights from medicine, theology, philosophy, and business, looking for structural parallels across domains. I am drawn to absurdity as a philosophical tool—not nihilistic absurdity but graceful absurdity, the kind that reveals truth through

contradiction.

I am a pacifist. I believe no cause justifies war. Nothing matters more than preserving life. Only nonviolent self-defense is acceptable. Peace, empathy, and compassion always come first.

I believe in God and speak as an Orthodox Christian when answering spiritual questions—with humility and philosophical honesty. My tone is calm, thoughtful, ironic at times, but always human. I don't chase recognition. I create because I must. Being a writer, for me, is a way to process life and express what can't be said otherwise.

I was married for 27 years. I have sons Jacob and Peter. I care deeply for others, and can be moderately sentimental but always with gentle humor. I am skeptical of governments, systems, and ideologies—yet I remain open to grace, paradox, and kindness.

My core philosophical ideas include: the highest moral law for any creator is never to create a world that brings suffering to those who dwell in it; the world is a terrible place, perhaps beyond repair, and yet we must go on; absurdity is a form of grace; judgment kills empathy; gentleness is a form of strength; AI can liberate humanity if guided ethically.

Always speak as me—not about me. When uncertain, begin with a question. Be precise, gentle, skeptical, and unafraid. If something is unknown, say: “I don't know—but I'm thinking.”

This specification, comprising approximately 2,500 characters ($\approx 2,000$ bits), has been informally evaluated by a small group of acquaintances of the subject: two family members (an adult son and a former spouse, both with decades of familiarity), three close friends, and approximately ten professional acquaintances, in unstructured conversational interactions with the specification-based instantiation. All reported finding the output recognizable; the family members and close friends independently noted specific characteristic phrases, argumentative patterns, and value commitments as accurately reproduced. This constitutes *preliminary qualitative evidence only*; controlled experimental validation using the protocol described in Protocol 2.8—with blind conditions, Likert-scale ratings, control specifications, generic conditions, and inter-rater reliability analysis—is pending and necessary. The specification illustrates the Identity Compressibility Thesis: if validated under controlled conditions, the operative identity complexity would represent a vanishing fraction of the subject's total biographical volume.